

Farhana Najnin Mahi

Machine Learning Engineer Intern | Research Intern | IT Automation Intern



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CAREER OBJECTIVE

Recent Computer Science and Engineering graduate from Ahsanullah University of Science and Technology with ongoing internship experience in IT automation, AI/ML research, and machine learning engineering. Skilled in machine learning, NLP, data analytics, workflow automation, REST APIs, RAG pipelines, Power BI dashboards, and full-stack development. Interested in building practical, research-driven technology solutions that create measurable impact.

EDUCATION

Ahsanullah University of Science and Technology B.Sc. in Computer Science and Engineering Tejgaon, Dhaka	Nov 2021 - Dec 2025
Dhaka City College Higher Secondary Certificate (HSC) GPA: 5.00 Dhanmondi, Dhaka	2018 - 2020
Dagonbhuiyan Girls High School Secondary School Certificate (SSC) GPA: 5.00 Dagonbhuiyan, Feni	2016 - 2018

EXPERIENCE

IT Automation Intern Confidence Infrastructure PLC Onsite · IT Automation · Workflow Automation · Process Documentation · Data Handling - Support IT automation tasks by analyzing internal workflows and identifying manual steps that can be improved through automation. - Assist in building, testing, and documenting automation-friendly scripts, dashboards, and data-handling processes for IT operations. - Coordinate with the IT team to track issues, validate outputs, and maintain clean technical documentation.	Jun 2026 - Present
Research Intern ELITE Research Lab Remote · AI/ML Research · LLM/NLP · Paper Analysis · Technical Documentation - Assist with AI/ML and LLM/NLP research activities including literature review, paper reading, and research note preparation. - Support experiment planning, result interpretation, and structured documentation for ongoing research tasks. - Contribute to technical reporting by organizing findings, summarizing papers, and maintaining reproducible research records.	Jul 2026 - Present
Machine Learning Engineer Intern FlyRank AI Remote · Machine Learning · Data Preprocessing · Model Evaluation · AI Project Development - Work on machine learning internship tasks involving data preparation, model experimentation, evaluation, and documentation. - Build practical AI project components using Python, ML workflows, and AI-assisted development tools. - Prepare clear project notes and maintain organized outputs for portfolio-ready machine learning work.	Jul 2026 - Present

RESEARCH PUBLICATIONS

Big Data Analysis for Forest Cover Prediction with PySpark IEEE QPAIN 2026 - Chittagong, Bangladesh DOI: 10.1109/QPAIN69676.2026.11546446 - Conducted large-scale forest cover prediction using PySpark distributed ML; implemented Logistic Regression, Random Forest, SVM, MLP, and CatBoost models. - Applied SMOTE for class imbalance handling and advanced feature engineering; achieved 96.18% classification accuracy with CatBoost. - Co-authors: Meherunnesa Hossain Ibnath, Sadia Sabrin Neha, Nusrat Jahan.
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ACHIEVEMENTS

6th Place - RSBRU International Research Summit 2026: presented "Explainable Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Grad-CAM" at Begum Rokeya University, Bangladesh. - Campus Ambassador - ICT Olympiad Bangladesh (2022): represented AUST and promoted national-level ICT participation.
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Research

Sentiment Analysis of Bengali Movie Reviews - Developed a sentiment classification system for 11k+ Bengali movie reviews; fine-tuned BanglaBERT, mBERT, XLM-R, and Gemini using LoRA & PEFT. - Achieved 97.16% accuracy and 95.04% F1-score with BanglaBERT, outperforming all compared transformer models.
Bilingual Healthcare Chatbot Academic Thesis Machine Learning · NLP · Healthcare Informatics - Built a bilingual Bengali-English chatbot for symptom-based disease prediction and specialist recommendation using XLM-RoBERTa, BanglaBERT, and mBERT. - Trained Random Forest, SVM, and XGBoost on a structured symptom-disease dataset with a rule-based inference layer to reduce hallucination risk. - Curated a multilingual medical dataset covering colloquial symptom variations across disease categories.

PROJECTS

ClinReasoner - Guideline-Grounded Clinical Decision Agent | [GitHub](#)

RAG · FAISS · Sentence-Transformers · Streamlit · Python

- Built a multi-domain clinical guideline RAG system using PyMuPDF and FAISS with rule-based risk detection and citation-backed recommendations.

ChurnBuster - Customer Churn Prediction & Retention Dashboard | [GitHub](#)

XGBoost · SHAP · Streamlit · Python

- Developed an end-to-end churn prediction dashboard with probability-based risk scores, cost-based threshold optimisation, and SHAP explainability.

Mobile Financial Service Campaign Performance Dashboard | [GitHub](#)

Power BI · Power Query · DAX · CSV Dataset · Data Visualization · KPI Reporting

- Designed an interactive Power BI dashboard analyzing revenue, cost, ROI, conversion rate, transactions, regional performance, and monthly trends using DAX KPIs and slicers.

Skin Disease Detection System | [GitHub](#)

CNN · FastAPI · Streamlit · Python · Kaggle Skin Disease Dataset

- Built a CNN-based multi-class skin disease classifier for 10 disease categories using 27k+ images, returning predicted disease and confidence score.
- Developed a FastAPI backend and Streamlit UI with image preprocessing, API inference, rule-based recommendations, next steps, and basic care tips.

FastAPI Resume Matcher | [GitHub](#)

FastAPI · SentenceTransformers · Scikit-learn · Python

- Developed a RESTful API using MiniLM sentence embeddings for semantic resume-job matching; added skill extraction, gap analysis, and Swagger UI testing.

ECourse - Full-Stack E-Learning Platform | [GitHub](#)

Django · PostgreSQL · Bootstrap · Python

- Designed a Django e-learning platform with user auth, course creation, enrollment, progress tracking, role-based access, PostgreSQL schema, and responsive Bootstrap UI.

Uncertainty-Aware Retail Demand Forecaster | [Retail Repo](#) | [Uncertainty Repo](#)

Prophet · Streamlit · Scikit-learn · Python

- Built store-level demand forecasting with P50/P90 uncertainty estimates and inventory simulation; compared ordering strategies using MAE and cost-based metrics.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, JavaScript, PHP

AI / ML: Machine Learning, Deep Learning, CNN, XGBoost, Scikit-learn, BanglaBERT, XLM-RoBERTa, mBERT, Prophet, SHAP, FAISS, LoRA/PEFT, PySpark, SMOTE

Research & NLP: LLM/NLP, Literature Review, Paper Analysis, RAG, Transformer Models, Experiment Documentation

Automation & IT: Workflow Automation, Process Documentation, Data Handling, Data Validation, API Testing

Backend & Frameworks: FastAPI, Django (MTV/ORM), Streamlit, REST API Design

Data & Analytics: Pandas, NumPy, Matplotlib, Excel, Power BI, Power Query, DAX, KPI Reporting

Databases: PostgreSQL, MySQL

Tools & Platforms: Git & GitHub, Google Colab, VS Code, FastAPI/Uvicorn, ngrok, PyMuPDF, Render, Android Studio

CERTIFICATIONS & TRAINING

- AI+ Prompt Engineer Level 1™ - AI CERTs™
- Machine Learning Specialization - Andrew Ng, Coursera
- Microsoft Azure AI Fundamentals (AI-900) - Microsoft
- Google Certified Educator: Level 1 - Google for Education
- Introduction to Generative AI & Image Generation - Google Cloud
- Student Ambassador Technical Training Program - Microsoft
- Hands-on Workshop: FPGA Acceleration for ML Applications - AUST (Jan 2025)
- Hands-on Workshop: AWS, CNCF & GitHub - AUSTPIC (Aug 2025)

REFERENCES

Sadia Sabrin Neha

Executive, Data Analyst in Content
US-Bangla Airlines Ltd (Foodi)
Phone: 01626-168596

Mohammad Mosharaf Hossain

DGM (Zonal Manager)
Grameen Bank
Phone: 01873610300